

The Mechanical Collaborator: Against the Anthropomorphization Trap in Human–AI Epistemic Inquiry

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Two temptations structure the current debate about AI-assisted research. The first is to anthropomorphize AI systems — to speak of them as believing, intending, committing, and collaborating in a sense continuous with human epistemic partnership. The second is to collapse all AI participation into instrumentation, treating AI outputs as the products of a passive tool that leaves the epistemic structure of inquiry entirely human. This paper argues that both temptations rest on the same false assumption: that collaboration either requires human-like epistemic agency or amounts to nothing at all. Against this, the paper defends *asymmetric epistemic collaboration* — a structure in which AI systems can make genuine, constitutive contributions to the development of an inquiry without possessing the normative properties of human epistemic agents. The paper’s central and counterintuitive claim is that the route to genuine collaboration runs *through* honest acknowledgment of the AI’s mechanical nature, not around it. The **Anthropomorphization Trap** — the attempt to justify collaborative language by inflating the AI’s epistemic status — is shown to undermine rather than establish the legitimacy of human–AI inquiry. A transparently mechanical participant, whose computational properties are known and whose interaction structure is documented, contributes more genuinely to research than a rhetorically humanized one. The paper engages directly with Mellette’s (2026) argument that human–AI collaboration is a category error, with Woodruff and Hewitt’s (2026) framework of epistemic agency limitations in large language models, and with Rodrigues’s (2026) Apt Curation Model. It draws on Huynh’s ongoing human–AI research corpus as a methodologically self-conscious instance of asymmetric collaboration in practice.

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1 Introduction

Generative AI has changed the conditions under which research is conducted, and it has not changed them neutrally. A researcher can now ask a system to generate competing hypotheses, reconstruct an argument from an unfamiliar theoretical tradition, identify counterexamples, expose inconsistencies, and propose distinctions that were absent from the researcher’s initial formulation. The inquiry may proceed differently — substantially differently — because the system produced these outputs. The question is how to describe what happened.

Two vocabularies are available, and neither is adequate.

The *tool vocabulary* says the researcher used an instrument. The AI executed operations specified in advance, and the researcher evaluated the results. The epistemic structure remained entirely human. This vocabulary captures the researcher’s continuing control over the inquiry and the AI’s inability to initiate, persist in, or take responsibility for a research project. But it becomes strained when the system’s outputs are not merely executed operations whose epistemic significance was fixed in advance — when they are, instead, contingent interventions that introduce new directions into a developing inquiry.

The *partner vocabulary* says the researcher collaborated with an AI. It captures the intuition that interaction with the system can alter the trajectory of inquiry in ways that feel substantively different from consulting a database or running a statistical package. But this vocabulary imports, without explicit endorsement, a cluster of assumptions about the AI’s epistemic properties — that it has opinions, that its agreement indicates genuine assent, that its persistence with a position reflects genuine commitment — which are false of current AI systems and which compromise the integrity of inquiry when treated as true.

The philosophical literature has recognized this impasse but not resolved it. Mellette (2026) argues that the collaborative vocabulary is a category error because responsibility cannot be genuinely distributed across a human–AI dyad. Woodruff and Hewitt (2026) argue that LLMs lack the reflective epistemic agency that genuine collaboration would require. Rodrigues (2026) proposes that the human occupies the role of virtuous curator rather than co-author, repositioning the question from collaboration to authorship. Each of these positions identifies something important. None of them dissolves the impasse.

A parallel literature has emerged at the level of research methodology rather than philosophical foundation. Sułkowski, Dacko-Pikiewicz, and Szczepańska-Woszczyńska’s *Epistemology and Methodology of Research in the AI Age* (2026) — the opening volume of an eight-volume doctoral-training series, *New Research Methodologies for PhD and Beyond* — maps how digital platforms, administrative systems, sensors, APIs, and large text and network datasets are reshaping the practical conduct of inquiry alongside surveys, interviews, and experiments. Its companion volume, *AI in Social Sciences Research* (Sułkowski et al., 2026), addresses applied technique directly. This literature is indispensable for training researchers to work responsibly with AI systems. But methodological guidance presupposes an answer to a question it does not itself ask: what kind of epistemic relation is being managed when a researcher works with an AI system? A volume can specify how to document an interaction without settling whether the documented interaction constitutes collaboration, curation, or mere instrumentation. That prior question is this paper’s target.

This paper argues that the impasse is sustained by a shared assumption: that there are only two positions — the tool and the human-like partner — and that the argument is about which of them is correct. The paper defends a third position. AI systems can be genuine collaborative participants in inquiry without being human-like epistemic agents. The conditions for this participation are: transparency about the AI’s mechanical nature, documentation of the interaction structure, and calibration of the AI’s epistemic authority to what its production process actually warrants.

The counterintuitive result is that *de-anthropomorphization is the condition for genuine collaboration, not its obstacle*. A researcher who honestly acknowledges the AI’s computational nature and manages the interaction accordingly is in a better position to integrate AI contributions legitimately than a researcher who inflates the AI’s status in order to justify using the language of collaboration. The Anthropomorphization Trap closes around the latter; the former escapes it.

The argument proceeds as follows. Section 2 reconstructs the current debate through three positions that together define the live disagreement. Section 3 develops the Anthropomorphization Trap in detail, identifying its two failure modes and explaining why both of them compromise epistemic integrity. Section 4 develops the positive account of asymmetric epistemic collaboration and specifies the mechanical condition as its entry requirement. Section 5 develops the documentary consequence — the demand for a genealogical record of AI interaction — and draws on Huynh’s ongoing research corpus as a methodologically explicit instance. Section 6 addresses three objections. Section 7 concludes.

2 Three Positions in the Current Debate

2.1 The Category Error View

The most direct challenge to collaborative language comes from Mellette (2026), who argues that describing human–AI interaction as “collaboration” is a category error rooted in a misunderstanding of epistemic responsibility. On Mellette’s account, responsibility is not a divisible property that can be apportioned across a human–AI dyad. It is a unitary normative relation constituted by undivided *justificatory authority*: the capacity to account for one’s claims in the space of reasons, to revise them in light of evidence and argument, and to bear the normative consequences of their failure.

AI systems do not possess justificatory authority. Their outputs are not produced by a process of reasoning that tracks evidence and responds to argument. They are generated by statistical regularities in training data, structured by prompt dynamics, and unrelated to any standing commitment to the proposition being generated. When a researcher allows an AI system to resolve uncertainty, stabilize interpretation, or recommend analytical direction, the researcher does not share responsibility with the AI. The AI’s contribution displaces the epistemic grounding for the researcher’s subsequent claims, leaving responsibility formally attributed to the human while its normative basis has been structurally eroded.

Mellette’s argument has significant force. He is right that responsibility cannot be naively distributed across a human–AI dyad. He is also right that the phenomenology of deliberation — the feeling of having thought through a problem — is not a reliable guide to its normative structure. A researcher who has outsourced the framing of a problem to an AI system without recognizing that framing as an outsourcing may believe, sincerely, that the conclusion is their own.

But Mellette’s argument establishes more than it intends to. The argument moves from: *responsible collaboration requires symmetric justificatory authority* to: *wherever justificatory authority is asymmetric, the collaborative label is inappropriate*. This inference is valid only if collaboration is defined by symmetric justificatory authority — which is precisely the assumption this paper questions. Among human researchers, authority and responsibility are regularly distributed unevenly. A principal investigator who bears final responsibility for a published claim that was substantially shaped by a postdoctoral researcher’s empirical work has not thereby failed to collaborate with the postdoc. The collaborator and the responsible party are different people, exercising different kinds of authority, and this is normal rather than pathological.

Mellette’s argument needs to be distinguished from a weaker version that this paper accepts: the claim that responsibility cannot be *transferred* to or *shared with* an AI system. The present account agrees. What it denies is the inference from the non-transferability of responsibility to the impossibility

of genuine collaboration.

2.2 The Epistemic Agency Limitation View

Woodruff and Hewitt (2026) offer a more nuanced position grounded in a careful analysis of what epistemic agency requires. Drawing on Peirce’s account of inquiry, Sellars’ concept of the space of reasons, and Stanovich’s tripartite model of cognition, they argue that knowledge production requires three cognitive levels: autonomous processes, algorithmic processes, and reflective control. LLMs display strong performance at the autonomous and algorithmic levels — pattern recognition, hypothesis generation, analogical reasoning, sustained simulation — but reflective control, which governs the application of epistemic norms to the management of one’s own cognitive processes, remains distinctively and necessarily human.

The consequence, on their account, is that LLMs should not be positioned as epistemic peers in inquiry but as *Knowledge-Building Partners* (KBPs): systems designed to support inquiry by generating patterns, proposing hypotheses, maintaining representations, and amplifying the cognitive functions that human researchers perform at the autonomous and algorithmic levels, while withholding any claim to epistemic authority at the reflective level.

Woodruff and Hewitt’s descriptive claims about LLMs are largely correct, and this paper accepts them. LLMs lack persistent commitment, cross-session consistency, normative sensitivity, and the kind of reflective regulation that would allow them to monitor and correct their own epistemic processes. These are genuine limitations, and they are precisely the limitations that make collaboration necessarily asymmetric.

Where the present paper diverges from Woodruff and Hewitt is in the normative conclusion they draw. They treat the KBP framework primarily as a design specification — a set of properties that AI systems should have in order to avoid epistemic overreach. The present paper argues that their framework implicitly describes a *collaborative role*. A KBP that generates hypotheses, maintains shared representations, produces candidate counterexamples, and sustains analytical threads across a long inquiry is not merely a tool. Its contributions are constitutive of the inquiry’s development — the inquiry would have followed a different trajectory without them — and the interaction has a reciprocal structure that distinguishes it from one-directional instrumental use.

Woodruff and Hewitt’s framework thus supports, rather than undermines, the account developed here. Their analysis explains why the collaboration must be asymmetric — why the AI cannot bear reflective epistemic responsibility — without establishing that asymmetric collaboration is impossible. On the contrary, their design requirements for the KBP specify what a mechanical epistemic participant should look like in order to contribute legitimately to inquiry.

2.3 The Apt Curation Model

Rodrigues (2026) develops a third position that reframes the question from collaboration to authorship. Drawing on Floridi’s concept of “distant writing,” Nozick’s distinction between originitive and constitutive contribution, and Sosa’s virtue epistemology, Rodrigues argues that the relevant question is not whether the researcher authored every sentence in the resulting text but whether they exercised the virtues of an *apt curator* of a text-generation process.

The Apt Curation Model (ACM) identifies four curatorial virtues: the *dialogical* virtue of engaging AI outputs critically and iteratively; the *evaluative* virtue of verifying, selecting, and rejecting generated content; the *integrative* virtue of synthesizing outputs into a coherent analytical structure; and the *architectural* virtue of designing the overall form of the inquiry. A researcher who exercises these virtues

in relation to AI-generated text is, on Rodrigues's account, genuinely the author of the resulting work — not because they produced every sentence but because they exercised the epistemic agency through which the text acquired its intellectual character.

The ACM is a genuine contribution, and the curatorial virtues Rodrigues identifies are real epistemic skills whose cultivation is practically urgent. The evaluative virtue — the capacity to assess AI outputs against standards appropriate to their source, to reject confabulation, to detect distributional bias, and to verify empirical claims — is precisely what distinguishes responsible human–AI inquiry from credulous receptivity.

The present paper's quarrel with the ACM is about scope. Rodrigues frames the model primarily as an account of *authorship attribution* — of who deserves credit for the resulting text. This is a narrower question than the one about collaboration. The collaborative question concerns the structure of the epistemic interaction that produced the text: what participants contributed, how their contributions shaped the inquiry's development, and how the human's epistemic responsibilities were maintained across the process. The ACM answers part of this question — it specifies what the human must do — but does not fully theorize what the AI contributes and why those contributions have the epistemic significance they do.

Without an account of why AI outputs are *apt materials* for curatorial virtue — why engaging with them critically is an epistemic practice rather than an arbitrary exercise — the ACM risks treating AI outputs as interchangeable raw materials for an essentially human process. The present paper argues that what makes AI outputs apt for critical engagement is precisely their nature as structured outputs of a characterizable mechanical process — which requires that the researcher understand that nature, rather than misrepresent it through anthropomorphizing language.

3 The Anthropomorphization Trap

3.1 The Structure of the Trap

The Anthropomorphization Trap arises from a genuine intellectual pressure. Researchers who work with AI systems across extended inquiries observe something real: the AI's contributions change the trajectory of their thinking in ways that feel substantively different from using a calculator or running a literature database search. The system generates objections the researcher had not considered, proposes distinctions that restructure the problem, identifies counterexamples that force substantive revision. Describing this as “using a tool” feels inadequate — not merely imprecise but methodologically dishonest.

The trap springs when researchers reach for the nearest available vocabulary for intellectually significant reciprocal interaction: the vocabulary of human collaboration. This vocabulary imports, without explicit endorsement, a cluster of assumptions about the AI's epistemic properties.

The AI has opinions: when it produces a claim, this reflects something like a belief, and its confidence signals something like epistemic warrant.

The AI is committed: when it persists with a position across an extended exchange, this reflects something like considered judgment rather than distributional artifacts.

The AI is responsive: when it revises a claim after criticism, this reflects something like genuine updating on evidence rather than prompt-conditioned output adjustment.

The AI is a peer: when it agrees with the researcher's conclusion, this constitutes something like independent corroboration.

All of these assumptions are false. And their falseness generates two distinct and serious failure modes.

3.2 Failure Mode One: Epistemic Inflation

When AI outputs are received as quasi-testimonial — as reports from an entity with genuine beliefs — the researcher adopts a posture of trust that is inappropriate to the source. The result is *epistemic inflation*: AI-generated claims acquire a credibility that their production process does not warrant.

A conclusion that the AI repeatedly produces is not more likely to be true because of that repetition; the repetition reflects distributional regularities in training data, not a convergence of independent reasoning toward a well-founded conclusion. A framing that the AI consistently applies is not more analytically apt because of that consistency; it may reflect training distributions that overrepresent certain analytical traditions or underrepresent alternative framings. A claim that the AI generates confidently is not more warranted than a claim it generates hesitantly; confidence is a feature of the generated text, not a reliable signal of the underlying reliability of the output.

Epistemic inflation is particularly dangerous in domains where the AI’s outputs are plausible and internally coherent but empirically unverified. The researcher who mistakes plausible coherence for warranted belief has incorporated false premises into the inquiry, and if those premises are not detected through external verification, they persist in the resulting claims.

3.3 Failure Mode Two: Accountability Evasion

The second failure mode is subtler. When the AI is treated as a collaborator in the human sense, the question of epistemic responsibility becomes confused. A researcher who believes they collaborated with an AI to develop a claim may feel that the claim emerged from a joint epistemic process — that responsibility is somehow distributed between the researcher and the system.

This is confusion masquerading as epistemic sophistication. The AI bears no responsibility for any claim in the resulting inquiry. Its outputs were generated by a computational process that did not weigh evidence, evaluate arguments, or commit to a conclusion. The human researcher remains the sole epistemic responsible party. But the language of collaboration can obscure this fact — not through any deliberate dishonesty but through the background assumption, imported with the vocabulary of human partnership, that responsibility is shared where inquiry is shared.

Huynh (2026a) identifies this as the central risk of what he calls “the borrowed rubric” — the application of evaluation standards from one epistemic context (peer collaboration) to a structurally different context (human–AI inquiry), which systematically misdescribes the normative structure of the second context by forcing it into the categories of the first.

3.4 Why the Trap Closes Both Ways

Both failure modes have generated a response in the philosophical literature: the recognition that anthropomorphized AI collaboration is epistemically irresponsible. This recognition is correct. But it has led some critics — Mellette (2026) most explicitly — to conclude that all collaborative language is therefore inappropriate for human–AI interaction.

This conclusion follows only if the only alternative to anthropomorphization is the strict tool view. But this is precisely the dichotomy the present paper rejects. The trap closes the other way when the recognition of anthropomorphization’s failure leads to the dismissal of all non-instrumental AI participation, rather than to the articulation of what non-anthropomorphized collaboration would look like.

The second jaw of the trap is the symmetry assumption: if genuine collaboration requires equivalent epistemic properties, and AI lacks those properties, AI cannot be a genuine collaborator. The assumption is the error. The present paper’s task is to articulate a form of genuine collaboration that does not require symmetrical epistemic properties — and to show that the condition for this form of collaboration is

precisely the acknowledgment of asymmetry.

4 Asymmetric Epistemic Collaboration and the Mechanical Condition

4.1 Separating Collaboration from Symmetric Agency

The existing debate conflates two questions that are conceptually distinct:

The participation question: Did this entity contribute to the development of the inquiry?

The responsibility question: Does this entity bear normative accountability for the resulting claims?

In ordinary human collaboration, these questions typically receive the same answer, because human collaborators normally both contribute and bear responsibility. The appearance of conceptual unity is an artifact of the human case: it reflects the fact that the properties of contribution and responsibility normally co-occur in human agents, not that they are conceptually identical.

Once this conflation is dissolved, the space for asymmetric epistemic collaboration opens. An entity can contribute to the development of an inquiry — can make contributions that are constitutive of the inquiry’s trajectory — without bearing normative responsibility for the resulting claims. A research assistant who contributes a literature review and a set of hypotheses contributes genuinely to the inquiry without bearing the principal investigator’s responsibility for the theoretical claims. The contribution is real; the responsibility is asymmetric.

The AI case extends this asymmetry further. The asymmetry is not merely institutional — a matter of professional hierarchy — but metaphysical: the AI lacks the kind of agency that ordinarily grounds epistemic responsibility. This is why the collaboration must be *asymmetric* rather than merely institutionally unequal. It does not follow that the AI’s contribution is not genuine. It follows that the normative endpoint of the collaboration is entirely human-controlled.

4.2 The Positive Account

An epistemic relation is collaborative when multiple participants make structured contributions that materially alter the development of a shared inquiry, where those contributions are integrated through *reciprocal epistemic activity*: the system’s outputs become objects of human evaluation, which generate further prompts, which produce further outputs, which are evaluated again.

This definition deliberately excludes mere causal influence. A search engine influences which sources a researcher encounters. An autocomplete system influences the phrasing of a sentence. A poorly heated office provokes a line of thought. None of these are collaborative contributions, because the interaction lacks the reciprocal structure: they do not produce outputs in response to the researcher’s epistemic activity, and they are not modified by the researcher’s critical engagement with their prior outputs.

The AI interaction has this reciprocal structure when it is used appropriately: the researcher asks the system to generate hypotheses, evaluates them, accepts some in modified form, rejects others, asks the system to attack the accepted hypotheses, evaluates the resulting objections, revises the hypotheses in response, and so on. This is not instrumental use in the narrow sense. It is an iterative epistemic interaction whose outcome — the trajectory of the inquiry — is jointly produced by the human’s evaluative decisions and the AI’s generative outputs.

The collaboration is genuine. The responsibility is asymmetric.

4.3 The Mechanical Condition

The paper’s central claim is that the condition for legitimate asymmetric collaboration is the *honest acknowledgment of the AI’s mechanical nature by the human participant*.

This claim is counterintuitive and requires careful statement. It might seem that collaboration is strengthened by attributing richer epistemic properties to the AI: the more the AI resembles a human partner, the more the collaboration resembles human collaboration. The argument here runs in the opposite direction.

When a researcher anthropomorphizes AI, they introduce epistemic unreliability into the collaborative process. They attribute properties to the AI’s outputs — authority from commitment, credibility from belief, reliability from judgment — that those outputs do not possess. The inquiry is built, in part, on misattributed properties. Even if the resulting claims are correct, they were reached through a process that incorporated systematically false premises about one participant’s contributions.

When a researcher treats the AI as a *characterizable mechanical participant* — a system whose outputs are generated by a trained model with known architectural properties, whose responses are sensitive to prompt structure in documentable ways, whose limitations include non-persistence, non-commitment, susceptibility to distributional patterns, and a structural incapacity for cross-session consistency — the researcher can integrate AI contributions accurately. The outputs are treated as what they are: materials for inquiry, generated by a process whose properties are understood, to be evaluated by standards appropriate to their source.

The mechanical condition has three components.

Transparency about the production process. The researcher must understand, at a functionally adequate level, that AI outputs are generated by pattern-matching on training data, shaped by prompt structure, and not constrained by commitment to consistency across sessions. Huynh (2026b) documents this structural feature — what he terms *statelessness* — as not merely a practical limitation but a philosophically significant property of the AI participant: the AI does not remember its prior contributions to the inquiry, cannot maintain a research commitment across sessions, and begins each interaction without access to the epistemic genealogy of the project. Understanding this is not optional for the researcher who takes the mechanical condition seriously.

Documentation of the interaction structure. Because the AI’s contributions are characterizable only by reference to the prompts that produced them, the collaborative process is epistemically auditable only if the interaction is recorded. What prompts were sent, what outputs were generated, which were accepted, which were rejected, which generated further investigation, and which were abandoned — this record constitutes what Huynh (2026c) calls the *epistemic genealogy* of the inquiry’s development. Without it, the collaborative structure is invisible behind the surface of the resulting text.

Calibration of the AI’s epistemic authority. The AI’s contributions must be received at the appropriate epistemic level: as hypotheses, framings, and candidate distinctions, not as conclusions. This is the evaluative virtue that Rodrigues (2026) identifies in his ACM, and the present paper accepts it — while adding that its precondition is an accurate understanding of what is being evaluated. A researcher who treats AI outputs as materials requiring rigorous critical engagement — precisely because they are generated by a process that does not itself perform that engagement — is in a better epistemic position than one who treats them as conclusions requiring only verification.

4.4 The Paradox Stated

The paradox can now be stated precisely.

The researcher who satisfies the mechanical condition — who treats the AI as what it is — is in

a better position to integrate AI contributions legitimately than the researcher who inflates the AI's epistemic status, for three reasons.

First, the mechanical researcher's epistemic responsibility is unambiguously demarcated. They know they are the sole evaluator and decision-maker in the inquiry. This clarity prevents the accountability evasion described in Section 3.3.

Second, the mechanical researcher evaluates AI outputs at the appropriate epistemic level. They are not misled into accepting a claim because the AI appeared confident, or trusting a framing because the AI consistently applied it. They treat AI outputs as materials — which is what they are.

Third, the mechanical researcher can document the collaborative process accurately, making the inquiry's development auditable and the AI's contribution legible without misrepresentation. This documentation is not merely a transparency measure. It is, as Section 5 develops, a constitutive feature of epistemically legitimate human–AI inquiry.

The upshot: *de-anthropomorphization is the condition for genuine collaboration, not its obstacle*. The anthropomorphized AI collaboration is not more collaborative for being anthropomorphized; it is less epistemically legitimate, because it rests on misattributed properties. The transparently mechanical collaboration is genuine precisely because it acknowledges what the AI is and integrates its contributions on that basis.

5 The Documentary Consequence: Genealogy as Epistemic Infrastructure

5.1 Why Documentation Is Not Optional

The mechanical condition generates a non-obvious consequence for research practice: the genealogical record of AI interaction is not merely an ethical requirement of transparency but an epistemic requirement of the inquiry's own integrity.

The reasoning is as follows. If the AI's contributions are constitutive of the inquiry's development — if the inquiry would have followed a different trajectory without them — then the inquiry's development is not fully visible in its outputs alone. The final text records the conclusions reached; it does not record how those conclusions were reached, which framings were considered and rejected, which hypotheses were introduced by the AI interaction and subsequently developed or abandoned, which distinctions entered the project through the prompt–response dynamic rather than through the researcher's independent reasoning.

This invisibility matters philosophically. In ordinary human inquiry, the process of reasoning is, at least in principle, recoverable from the text: the argument structure, the evidence cited, the objections addressed. In human–AI inquiry under the mechanical condition, an additional layer of the reasoning process — the generative layer — is contributed by a participant whose contributions are not preserved in the conventional textual record unless the interaction is explicitly documented.

Huynh (2026c) formalizes this as the *act-level criterion problem*: the question of whether a given AI contribution was constitutive of the inquiry cannot be settled by examining the resulting papers alone, because the papers record conclusions rather than the process of their development. The genealogical record — what was generated, what was selected, what was rejected, and what was subsequently revised — is the evidence for collaboration that the paper's surface cannot supply.

This demand converges with a consensus emerging outside philosophy proper. Sułkowski, Ratajczak, and Szczepańska-Woszczyna's *Research Ethics, Integrity, and Transparency* (2026) and Sułkowski, Woźniak, Seliga, and Lis's *Data Management and Digital Infrastructure in Social Sciences* (2026) —

companion volumes in the doctoral-methods series cited in Section 1 — independently arrive at documentation and data-infrastructure requirements for AI-assisted research, approached from research ethics and data management rather than from epistemology. The convergence is evidence that the genealogical requirement is not a philosopher’s invention but a structural feature of what responsible AI-assisted inquiry actually requires, approached from two directions at once.

5.2 The Research Sandbox as a Model

Huynh’s ongoing human–AI research programme — comprising the Longevity Asymmetry Corpus (LAC) and the Phylogenetic Closure Corpus (PCC), documented across 162 timestamped preprints — constitutes a methodologically self-conscious instance of asymmetric epistemic collaboration in practice. The programme explicitly adopts the mechanical condition: the AI is not positioned as a co-author or peer investigator but as a generative and analytical participant whose outputs require systematic human evaluation, selection, and integration.

The *research sandbox model*, documented in Huynh (2026d), treats the AI’s contributions as materials for a human-governed inquiry rather than as testimony from an epistemic partner. The AI generates hypotheses, proposes formalizations, constructs counterexamples, and maintains representational threads that the researcher evaluates, rejects, modifies, and builds upon. The resulting papers and revisions are preserved as timestamped artifacts recording not only the inquiry’s conclusions but its intellectual sequence: what was proposed, when, how formulations changed, which claims were subsequently narrowed or expanded, which research directions emerged from earlier experiments.

This archival function is a direct consequence of the mechanical condition. Because the AI cannot remember across sessions — its statelessness, analyzed in Huynh (2026b), is a structural rather than incidental feature of the participant — the human researcher bears full responsibility for maintaining the inquiry’s continuity. Rather than treating this as a deficiency to be minimized, the sandbox model treats it as a feature that is epistemically managed: the researcher provides relevant context at the beginning of each session, maintains the project’s intellectual genealogy independently, and treats the AI’s fresh-start position as a characterized property of the collaboration.

Huynh (2026b) notes a further consequence of statelessness that the mechanical condition converts from liability to asset: because the AI begins each session without the accumulation of prior framing pressures, it may generate outputs that a persistent system — one that had accumulated the inquiry’s own framings — might not produce. The stateless AI can, in this sense, serve as an adversarial generator that is structurally resistant to the inquiry’s own confirmation tendencies, provided the researcher does not supply the prior framings uncritically at the session’s opening.

5.3 Division of Labor and Epistemic Dependence

Huynh (2026e) extends Kitcher’s (1990) account of the division of cognitive labor to the case of human–AI research collaboration, with results that support the asymmetric account developed here. Kitcher’s framework describes how an epistemic community distributes its investigative labor across a problem space in ways that maximize collective coverage. The framework was developed for communities of human researchers — each with independent cognitive resources, resistant to common-cause bias from a shared training distribution.

In human–AI collaboration, the division of labor takes a distinctive form. The AI participant can cover vast areas of a problem’s surface at speed, generating candidate hypotheses and analytical framings with a breadth that no individual human researcher could match in the same time. But this coverage is stochastic, its depth unreliable, and its outputs cannot be trusted as testimony in Hardwig’s (1985) sense:

they are not produced by a process that involves commitment, consistency-tracking, or sensitivity to the distinction between what the AI knows and what it is confabulating.

The upshot is a division of labor in which the AI occupies the *generative and exploratory* position — surveying the problem’s surface, producing candidate structures, testing the logical consequences of proposed distinctions — while the human occupies the *reflective and evaluative* position — selecting which candidates to pursue, assessing their quality, integrating them into a developing argument, and bearing responsibility for the resulting claims. Neither position alone is sufficient for the inquiry. The division is not merely pragmatic. It is the structure that makes asymmetric epistemic collaboration epistemically legitimate rather than epistemically irresponsible.

Huynh (2026e) also extends Hardwig’s analysis of warranted trust in testimony to the AI case, arguing that the conditions for warranted trust — that the testifier have relevant evidence, that they communicate it reliably, and that the recipient have good reason to believe both — are not met for AI outputs received uncritically. This is not an argument against using AI outputs but an argument for using them *at the appropriate epistemic level*: as candidate materials requiring human verification and critical integration, not as testimony warranting default trust.

6 Objections

6.1 The Causation Objection

The most immediate objection is that the account conflates causal influence with collaboration. Every event that changes an inquiry’s trajectory is causally significant, but causal significance does not entail collaborative participation. A book encountered by chance on a train changes the research project. An autocomplete suggestion shapes a sentence. A serendipitous encounter with an unfamiliar thinker redirects the inquiry’s framing. None of these are collaborators; their causal contributions do not constitute epistemic participation.

This objection correctly identifies that causal influence alone is insufficient. The distinction between a causal event and a collaborative contribution is relational: it turns on whether the contribution was produced through a structured epistemic interaction with the reciprocal form described in Section 4.2.

The book on the train does not respond to prompts. The autocomplete suggestion is not modified in response to the researcher’s critical engagement with its prior suggestions. The serendipitous encounter generates a single causal influence, not an iterated epistemic interaction in which each round modifies the inputs for the next. The AI interaction, when governed by the mechanical condition, has exactly this iterated reciprocal structure: the researcher prompts, evaluates, modifies their prompting in light of the output, generates a new prompt incorporating the evaluation’s result, and so on. This structure is what distinguishes collaboration from causal influence — and it is a structural feature of the interaction, not a metaphysical property of the AI.

The mechanical condition matters here precisely because the distinction between causal event and collaborative participant is not inherent to the AI but produced by how it is used. An AI used to generate a one-shot output that is incorporated without critical engagement is functioning as a causal event — a sophisticated one, but causal rather than collaborative. An AI used iteratively, with critical evaluation at each step, whose outputs shape the next prompt and whose participation alters the trajectory of inquiry over time, is functioning as a collaborative participant. The mechanical condition specifies what the human must do to maintain the latter relationship legitimately.

6.2 The Responsibility Objection

Mellette's (2026) argument returns in a more targeted form. Even if the distinction between causal event and structured participant is granted, the asymmetry in epistemic responsibility is so extreme — the AI bears none at all — that calling the relation collaborative stretches the concept beyond recognition. Collaboration, in any philosophically significant sense, requires that all parties be accountable, in some sense, for the joint outcome. The AI is not accountable in any sense. The term “collaboration” therefore misdescribes the relation.

The objection correctly identifies that the AI bears no epistemic responsibility. The present account agrees. The question is whether this absence of responsibility eliminates the possibility of collaborative participation, or whether it establishes the specific character of the participation as asymmetric.

The relevant comparison is not with a symmetric human collaboration in which all parties share equivalent responsibility, but with a human collaboration in which responsibility is already unevenly distributed. Consider a case in which a senior researcher develops a theoretical position through sustained engagement with a junior researcher's empirical work, explicitly crediting that work as constitutive of the theory's development while bearing sole responsibility for the theoretical conclusions. The junior researcher's contribution is genuine — constitutive of the theory — even though the responsibility for the theory is entirely the senior researcher's. The asymmetry of responsibility does not retroactively render the junior researcher's contribution non-collaborative.

The AI case extends the asymmetry further and changes its grounds. But the extension and the change in grounds do not establish that collaboration has disappeared. They establish that the collaboration is asymmetric in a more radical way — that the human's responsibility is not merely greater than the AI's but is the only responsibility there is. This is precisely what the asymmetric account claims.

What the mechanical condition adds to this picture is the demand that the human's unilateral responsibility be clearly acknowledged and maintained throughout the inquiry. The researcher who satisfies the mechanical condition does not mistake the AI's contribution for shared responsibility. They know they are the sole responsible party. This knowledge — institutionalized in the documentation and calibration requirements — is what makes the collaboration epistemically legitimate rather than epistemically confused.

6.3 The Inflation Objection

A final objection targets the concept of collaboration from the other direction. If “asymmetric epistemic collaboration” describes a relation in which one party contributes without believing, intending, committing, or bearing responsibility, then the concept has been diluted to the point where it describes almost any sufficiently structured causal influence. The term “collaboration” retains its rhetorical weight while losing its analytical content.

This objection has force as a concern about concept maintenance. It is correct that the asymmetric account extends the concept of collaboration beyond its ordinary application domain. The question is whether this extension is warranted by the phenomenon or whether it represents an illegitimate inflation of the concept to cover cases it was never designed to describe.

The paper's response is that the concept's ordinary application domain — human researchers working together on a shared inquiry — already contains the structure that makes the AI case a genuine extension rather than an unwarranted inflation. That structure is: multiple participants making contributions that are constitutive of a shared inquiry's development, through a reciprocal epistemic interaction. The ordinary human case instantiates this structure with symmetric normative properties (commitment, responsibility, persistence). The AI case instantiates it without symmetric normative properties. The

structure is the same; the normative distribution is different.

The concept of collaboration in the asymmetric case does less work than in the symmetric case, because it cannot assign responsibility to the AI participant. But it does non-trivial work: it identifies the AI's contributions as constitutive of the inquiry's development (rather than merely causal), it demands documentation of the interaction structure, and it imposes obligations on the human participant — calibration, critical evaluation, genealogical recording — that pure instrumentation does not. This is meaningful content, not empty rhetoric.

7 Conclusion

The debate about human–AI epistemic collaboration has been constrained by a false dichotomy between the tool and the human-like partner. This paper has argued that both positions are errors, and that both rest on the same background assumption: that there is nothing between them.

The Anthropomorphization Trap occupies the space between the two positions, catching researchers who observe something real — the substantive, trajectory-altering contribution of extended AI interaction — and reach for the nearest available vocabulary, which imports false premises about the AI's epistemic properties. The trap closes in two ways: anthropomorphized collaboration compromises epistemic integrity; and the recognition of this compression leads critics to dismiss all collaborative language, returning to the tool view the evidence has already strained.

The exit from the trap is not a middle position between the tool and the partner. It is a different structure altogether: the mechanical participant, contributing genuinely to inquiry not by resembling a human epistemic agent but by being used as what it is — a characterizable mechanical system whose outputs are generated by a known process, integrated by a documented and critically governed interaction, and evaluated by standards calibrated to their source.

Three engagements with the existing literature have clarified the structure. Mellette (2026) is right that responsibility cannot be distributed across a human–AI dyad, but wrong that this eliminates the possibility of genuine collaboration. Woodruff and Hewitt (2026) are right that LLMs lack the reflective epistemic agency required for symmetric collaboration, but their framework of the Knowledge-Building Partner implicitly describes a genuine collaborative role — asymmetric, bounded, but real. Rodrigues (2026) is right that the human's curatorial virtues are genuine epistemic practices, but incomplete in not theorizing what makes AI outputs apt materials for curatorial engagement.

The positive account — asymmetric epistemic collaboration under the mechanical condition — specifies what makes a human–AI epistemic relation genuinely collaborative rather than merely instrumental, without requiring that the AI possess properties it does not have. The mechanical condition's three demands — transparency about the production process, documentation of the interaction structure, and calibration of epistemic authority — constitute the conditions of legitimacy for the collaborative relation.

The documentary consequence of this account is not optional. A human–AI inquiry that satisfies the mechanical condition but preserves no genealogical record of the interaction produces claims whose epistemic foundation is invisible. The inquiry may be epistemically legitimate in practice while being epistemically unauditible in principle — a condition that standard scholarly norms appropriately find inadequate.

Huynh's ongoing research corpus demonstrates that these conditions can be satisfied in practice. The sandbox model — iterative, documented, transparent about the AI's mechanical nature, asymmetric in responsibility throughout — is not a second-tier approximation of genuine inquiry. It is a distinctive form of epistemic practice with its own standards, its own division of labor, and its own conditions for

legitimacy. Articulating those conditions is the philosophical task that the proliferation of AI-assisted research has made urgent.

The philosophical question is not whether AI is secretly a human collaborator, but what collaboration means once the mechanical participant is among us.

That question remains inadequately answered.

This paper is a contribution toward answering it.

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